

Electrolyte Solutions — PCB Refurbishment App

Project Build Tracker | React Native + PostgreSQL | v1.0

Phase	Focus Area	Key Screens	Total Tasks	Done	% Complete	Status
Phase 1	Phase 1 — Database & Auth	Schema, RBAC, JWT	28	0	0%	Not Started
Phase 2	Phase 2 — Stock Module	Stock Summary, Inward	24	0	0%	Not Started
Phase 3	Phase 3 — Repair Pipeline	14-step repair flow, barc	32	0	0%	Not Started
Phase 4	Phase 4 — Dashboard & Analytics	Pending count, daily act	22	0	0%	Not Started
Phase 5	Phase 5 — Employee & Performance	Leaderboard, scoring, r	18	0	0%	Not Started
Phase 6	Phase 6 — Deploy & QA	App store, testing, secu	20	0	0%	Not Started
TOTAL			144	0	0%	Not Started

i Status auto-updates as you tick tasks Done in each checklist sheet. Use F5 key to force recalculate.

Phase 1 — Database, Auth & Infrastructure

Phase 1 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
PostgreSQL Schema Design							
1	DB Schema	Create `users` table	Fields: id, name, email, password_hash, role ENUM	To Do			
2	DB Schema	Create `lots` table	Fields: id, lot_no, batch, pixel_pitch, client_id,	To Do			
3	DB Schema	Create `panels` table	Fields: id, lot_id, sr_no, side ENUM(Left/Right), barcode	To Do			
4	DB Schema	Create `repair_steps` table	14 fixed steps: Panel Assign, Repair Aging, Panel	To Do			
5	DB Schema	Create `panel_logs` table	Fields: id, panel_id, step_id, engineer_id, timestamp,	To Do			
6	DB Schema	Create `defect_codes` table	Fields: id, code, description, category	To Do			
7	DB Schema	Create `performance_scores` table	Fields: id, user_id, month, year, pcbs_repaired,	To Do			
8	DB Schema	Create `clients` table	Fields: id, name, contact, email — link to lots	To Do			
9	DB Schema	Write all DB migration scripts (SQL)	Use sequential numbered migrations:	To Do			
10	DB Schema	Add indexes on frequently queried columns	lot_id, panel_id, engineer_id, step_id,	To Do			
Authentication & RBAC							
11	Auth	Set up JWT-based authentication	Access token (15min) + refresh token (7 days),	To Do			
12	Auth	Implement login endpoint	POST /auth/login — returns JWT on valid credentials	To Do			
13	Auth	Implement refresh token endpoint	POST /auth/refresh — rotates tokens	To Do			
14	Auth	Implement logout endpoint	POST /auth/logout — blacklist refresh token	To Do			
15	Auth	Build role middleware	Middleware validates JWT and injects req.user with	To Do			
16	RBAC	Define permission matrix per role	Superadmin: all. Manager: approve, view all. Team	To Do			
17	RBAC	Implement Row-Level Security in Postgres	Employee queries auto-filter to their own rows via RLS	To Do			
18	RBAC	Protect all API routes with role guards	Every endpoint must check minimum required role	To Do			
Environment & Infra Setup							
19	Infra	Set up PostgreSQL on server (or Supabase)	Recommended: Supabase for managed Postgres with	To Do			
20	Infra	Set up Node.js + Express API server	Or use Supabase Edge Functions — decide based	To Do			
21	Infra	Configure environment variables (.env)	DB_URL, JWT_SECRET, JWT_REFRESH_SECRET,	To Do			

Phase 1 — Database, Auth & Infrastructure

Phase 1 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
22	Infra	Set up CORS policy	Allow only app domains/origins	To Do			
23	Infra	Set up API rate limiting	Use express-rate-limit — protect login endpoint	To Do			
24	Infra	Set up logging (Winston or Pino)	Log all API requests and errors to file + monitoring	To Do			
25	Infra	Initialize React Native project (Expo)	npx create-expo-app ElectrolyteSolutions --	To Do			
26	Infra	Configure Expo EAS build	eas.json for development, preview, production profiles	To Do			
27	Infra	Connect frontend to backend (Axios + interceptors)	Auto-inject JWT token in every API request header	To Do			
28	Infra	Set up Git repo with branch strategy	main (prod), develop, feature/* branches	To Do			

Phase 2 — Stock Module (Inventory Management)

Phase 2 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
Stock Summary Screen (as seen in mockup)							
1	UI	Build Stock Summary page layout	Matches mockup: header with	To Do			
2	UI	Summary KPI cards: Total Lots, Received, Dispatched, Available	Cards with large numbers and color coding	To Do			
3	UI	Build Lot table with columns	LOT NO, BATCH, PIXEL PITCH, RECEIVED,	To Do			
4	UI	Status badge: Complete (green), In Process (amber)	Pill-style badges matching mockup colors	To Do			
5	UI	Action buttons per row: History icon, Export icon, Toggle (active/inactive)	3 action icons per lot row	To Do			
6	UI	Lot table pagination or infinite scroll	For lots with many rows	To Do			
7	UI	Export Lot + Export All buttons	Export to CSV/Excel, requires manager+ role	To Do			
Inward / Outward / Return / Redispatch Forms							
8	Form	Inward form: add new lot	Fields: Lot No, Batch, Pixel Pitch, Client, Received Qty	To Do			
9	Form	Outward form: log dispatch from a lot	Fields: Lot No (dropdown), Qty dispatched — updates	To Do			
10	Form	Return form: log return from client	Fields: Lot No, Qty returned, Reason — updates return	To Do			
11	Form	Redispatch form	Fields: Lot No, Qty — updates redispatch count	To Do			
12	Form	GRN discrepancy validation	If received qty ≠ expected: show warning and require	To Do			
13	Form	Auto-calculate Available = Received - Dispatch + Return - Redispatch	Formula must be server-side, not client-side only	To Do			
14	Form	Auto-set lot status to Complete when Available = 0	Trigger on each transaction update	To Do			
Lot History & Audit							
15	Audit	Lot history modal/page: full transaction log per lot	Show all inward/outward/return/redis	To Do			
16	Audit	Lock lot editing once marked Complete	Only superadmin can unlock a completed lot	To Do			
17	Audit	Prevent duplicate lot numbers	Validation at API level: unique constraint on lot_no	To Do			
18	Audit	Log all stock changes with actor (who did what)	Required for anti-fraud / data integrity	To Do			
19	Audit	Build API endpoints: GET /lots, POST /lots/inward, POST /lots/outward, etc.	Full CRUD with role-check middleware	To Do			
20	Audit	Filter by client, lot status, date range	Server-side filtering for all lot list queries	To Do			
21	Audit	Barcode/lot number search in stock view	Search bar matches mockup UI	To Do			

Phase 3 — Repair Pipeline (14-Step Process)

Phase 3 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
Repair Form — Panel Assignment (Step 1)							
1	Repair	Build Repair page layout	Matches mockup: Find Panel section, Panel	To Do			
2	Repair	Find Panel: search by barcode, lot, SR no, engineer	Search bar returns panel details instantly	To Do			
3	Repair	Panel Information display: Lot No, SR No, Side, Barcode	4 info cards shown at top once panel found	To Do			
4	Repair	14-step progress strip (visual pipeline)	Numbered steps across top — active step highlighted in	To Do			
5	Repair	Step 1 — Panel Assign form	Fields: Lot No dropdown, SR No, Side (Left/Right),	To Do			
6	Repair	Barcode auto-generation logic	Format: ESRP2P5925E26128R0341	To Do			
7	Repair	Assign Panel button — POST to API	Validates required fields before submitting	To Do			
8	Repair	Next button advances to next step	Only enabled after current step is successfully saved	To Do			
All 14 Repair Steps — Forms & Logs							
9	Repair Steps	Step 2 — Repair Aging: log engineer + start time	Record who started aging and timestamp	To Do			
10	Repair Steps	Step 3 — Panel Opening: log engineer + remarks	Optional defect observation field	To Do			
11	Repair Steps	Step 4 — Silicon Removing: log engineer + count completed	Group by Lot view (as in mockup modal)	To Do			
12	Repair Steps	Step 5 — IC Removing: log engineer + defect code	Dropdown for defect type	To Do			
13	Repair Steps	Step 6 — IC Cleaning: log engineer + method used	Track cleaning method for quality audit	To Do			
14	Repair Steps	Step 7 — IC Replacing: log engineer + IC part used	Part code field	To Do			
15	Repair Steps	Step 8 — Debugging: log engineer + issue found + resolution	Text field for debug notes	To Do			
16	Repair Steps	Step 9 — 1st Aging: log engineer + duration	Track aging duration	To Do			
17	Repair Steps	Step 10 — Applying Silicon: log engineer + quantity used	Numeric field	To Do			
18	Repair Steps	Step 11 — Half Fitting: log engineer	Completion timestamp only	To Do			
19	Repair Steps	Step 12 — Mesh Fitting: log engineer + defect if any	Optional defect field	To Do			
20	Repair Steps	Step 13 — QC: log QC engineer + pass/fail decision	If fail → send back to relevant step; log which	To Do			
21	Repair Steps	Step 14 — Dispatch: log dispatch engineer + final qty	Mark panel as dispatched; update lot available count	To Do			
22	Repair Steps	All steps: prevent skipping — enforce sequential order	API must reject out-of-order step submissions	To Do			

Phase 3 — Repair Pipeline (14-Step Process)

Phase 3 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
Repair Step Detail Modal (as in mockup)							
23	Modal	Step detail modal: show all panels in a step grouped by lot	As per Silicon Removing mockup: Barcode, SR No,	To Do			
24	Modal	Search/filter inside modal by barcode, lot, engineer	Search bar inside modal	To Do			
25	Modal	Group by Lot toggle in modal	Toggle switch in modal header (as seen in mockup)	To Do			
26	Modal	Show panel count in modal header (e.g. 66 panels)	Count updates with filter	To Do			
27	Repair	Debug History screen	Button in repair header → shows all debug logs with	To Do			
28	Repair	Task Manager screen	Shows all panels currently assigned to the logged-in	To Do			
29	Repair	Bottleneck detection: flag steps where queue > threshold	Alert on dashboard when a step has too many pending	To Do			

Phase 4 — Dashboard & Analytics

Phase 4 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
Pending Count Section (as in mockup)							
1	Dashboard	Build Pending Count section	4 KPI cards: In Pipeline, Dispatched, Steps with	To Do			
2	Dashboard	Filter by Lot dropdown	Shows All Lots by default, then Lot 18/19/20 etc.	To Do			
3	Dashboard	Step cards: show pending count per step per lot	Each step card shows step number, name, lot, pending	To Do			
4	Dashboard	Clear filter button	Resets to All Lots view	To Do			
5	Dashboard	View Details button → opens pending detail modal	Orange button, opens modal with full pending	To Do			
6	Dashboard	Pending detail modal: table with SR No, Side, Lot No, Batch, Pending Step	Matches mockup exactly; left/right side counts shown	To Do			
7	Dashboard	All Steps sidebar in modal: grouped by step with count badges	Step list on left with pending count per step	To Do			
8	Dashboard	Lot tab filter inside modal (e.g. All Lots / Lot 18 (48))	Tab bar at top of modal	To Do			
9	Dashboard	Search Sr No. in modal	Search field in modal header	To Do			
Daily Process Count Section							
10	Dashboard	Daily Process Count section	Shows: Total Activity, Peak Step, Bottlenecks	To Do			
11	Dashboard	Date picker: Today / Yesterday / Custom date	3 options with calendar picker (matches mockup)	To Do			
12	Dashboard	Refresh button for real-time update	Manual refresh trigger	To Do			
13	Dashboard	Step-wise bar chart / progress bars	Each of 14 steps has a row with activity count and vs-	To Do			
14	Dashboard	vs prev indicator per step	e.g. ↑-50 vs prev — color coded red/green	To Do			
15	Dashboard	Bottleneck badge: auto-detect steps with most pending	Orange/red alert badge on bottleneck step	To Do			
Charts & Trend Views							
16	Analytics	Line chart: cumulative panel activity Feb–May	As seen in overview chart in mockup	To Do			
17	Analytics	Filter chart by date range	Month/quarter/custom range	To Do			
18	Analytics	Engineer performance chart	Bar chart: top engineers by panels processed per	To Do			
19	Analytics	Defect frequency chart	Pareto chart of defect codes	To Do			
20	Analytics	Lot progress chart	Per-lot breakdown: received vs dispatched vs available	To Do			
21	Analytics	Export dashboard data to Excel	Manager/superadmin only	To Do			

Phase 4 — Dashboard & Analytics

Phase 4 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
22	Analytics	Auto-refresh dashboard every 5 mins	Configurable interval	To Do			

Phase 5 — Employee Management & Performance

Phase 5 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
Employee Management							
1	Employees	Employee list page (manager/superadmin view)	Table: Name, Role, Line, Attendance %, Active	To Do			
2	Employees	Add / edit employee form	Fields: Name, Email, Password, Role, Line	To Do			
3	Employees	Employee profile page	Shows personal stats: PCBs repaired, avg time,	To Do			
4	Employees	Activate / deactivate employee toggle	Deactivated employees cannot log steps	To Do			
5	Employees	Line assignment: assign employee to production line	Line A, B, C etc.	To Do			
Performance Scoring System							
6	Scoring	Implement scoring formula	Score = (PCBs Repaired × 35%) + (First-pass Yield ×	To Do			
7	Scoring	Calculate PCBs repaired per employee per month	COUNT of panels where engineer_id = user and step	To Do			
8	Scoring	Calculate first-pass yield per employee	Panels completed without returning to a prior step /	To Do			
9	Scoring	Calculate avg time per PCB per employee	Total time from step 1 to step 14 per panel, average	To Do			
10	Scoring	Calculate attendance %	Manual entry by Team Lead or HR — daily check-in log	To Do			
11	Scoring	Auto-run monthly score calculation on 1st of each month	Cron job / scheduled function	To Do			
12	Scoring	Store monthly scores in performance_scores table	Historical record — never overwrite previous months	To Do			
Leaderboard & Rewards							
13	Leaderboard	Monthly leaderboard screen	Shows all employees ranked by score — highlight	To Do			
14	Leaderboard	Top 3 reward badge on leaderboard	Visual indicator + optional push notification	To Do			
15	Leaderboard	Employee self-view: my rank + my score breakdown	Employee can only see their own detailed breakdown	To Do			
16	Leaderboard	Manager view: full leaderboard with score breakdown per employee	Drill-down per employee	To Do			
17	Leaderboard	Historical leaderboard: view past months	Month/year selector	To Do			
18	Leaderboard	Competition mode: show team vs team (line vs line)	Optional — toggle between individual and team	To Do			

Phase 6 — QA, Security & App Store Deployment

Phase 6 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
Testing							
1	QA	Unit tests for all API endpoints	Use Jest + Supertest — test every route with valid and	To Do			
2	QA	Test role-based access: verify employee cannot access manager routes	Automated role guard tests	To Do			
3	QA	Test barcode uniqueness — no duplicate barcodes across lots	Edge case: same SR no in different lots must still be	To Do			
4	QA	Test sequential step enforcement — cannot skip steps	Try submitting step 5 before step 4 completes	To Do			
5	QA	Test GRN discrepancy alert logic	Submit batch where received < expected; verify	To Do			
6	QA	Test monthly score calculation correctness	Manual calculation vs system output — verify for 5	To Do			
7	QA	UAT with actual factory floor operators	Run 1-week parallel pilot: paper logs + app	To Do			
8	QA	Performance test: 50 concurrent users logging steps	Ensure API responds < 500ms under load	To Do			
Security & Data Integrity							
9	Security	Validate all inputs server-side (not just client)	Use Zod or Joi for request body validation on every	To Do			
10	Security	Hash passwords with bcrypt (rounds ≥ 12)	Never store plain text; never log passwords	To Do			
11	Security	Protect against SQL injection (use parameterized queries)	No raw string interpolation in SQL	To Do			
12	Security	Implement audit log: who changed what, when	Every POST/PUT/DELETE logs actor + before/after	To Do			
13	Security	HTTPS enforced on all API endpoints	Redirect HTTP → HTTPS	To Do			
14	Security	Implement data backup: daily automated PostgreSQL dump	Backup to separate storage (S3 or equivalent)	To Do			
App Store Deployment							
15	Deploy	Configure Expo EAS build for Android APK	eas build --platform android	To Do			
16	Deploy	Configure Expo EAS build for iOS IPA	eas build --platform ios (requires Apple Developer	To Do			
17	Deploy	Create Google Play Store listing	App name, description, screenshots, privacy policy	To Do			
18	Deploy	Create Apple App Store listing	Same content — plus Apple review compliance	To Do			
19	Deploy	Set up CI/CD pipeline	GitHub Actions: run tests → build → deploy to stores on	To Do			
20	Deploy	Set up crash reporting (Sentry)	Catch and alert on production app errors	To Do			
21	Deploy	Configure push notifications (Expo Notifications)	For leaderboard reward alerts and step assignment	To Do			

Phase 6 — QA, Security & App Store Deployment

Phase 6 Detailed Checklist — Electrolyte Solutions PCB App

#	Category	Task / Deliverable	Notes / Detail	Status	Owner	Due Date	Notes
22	Deploy	Deploy API to production server	Railway, Render, or VPS — configure environment	To Do			